

OpenWPM – Installation, Configuration & Validation Report

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Environment: Kali Linux (VirtualBox)

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1. Introduction

OpenWPM (Open Web Privacy Measurement) is an open-source framework designed to automate web browsing at scale and collect detailed measurements related to **web tracking, cookies, fingerprinting techniques, third-party requests, and other privacy-impacting behaviors**.

It uses real browser instrumentation (Firefox + Selenium) along with a custom browser extension to provide high-fidelity measurements of website behavior. OpenWPM is widely used in **privacy research, cybersecurity assessments, compliance audits, and web measurement studies**.

This report documents the **complete end-to-end installation, configuration, troubleshooting, and validation of OpenWPM** in a Kali Linux environment. The goal is to provide a **reproducible, beginner-friendly, and mistake-free setup guide** so that another engineer can set up OpenWPM later without ambiguity.

2. Use Cases

OpenWPM can be used for :

- Web privacy and tracking audits
- Cookie and third-party tracker analysis
- Browser fingerprinting research
- Compliance validation (GDPR / DPDP / cookie consent analysis)
- Security research on ad-tech and tracking ecosystems
- Automated crawling and website behavior monitoring
- Academic and enterprise privacy measurement projects

3. System Requirements

Component	Minimum Requirement
CPU	2 cores (4+ recommended)
RAM	4 GB (8+ GB recommended)
Storage	20 GB free disk space
Network	Stable internet connection

4. Software Requirements

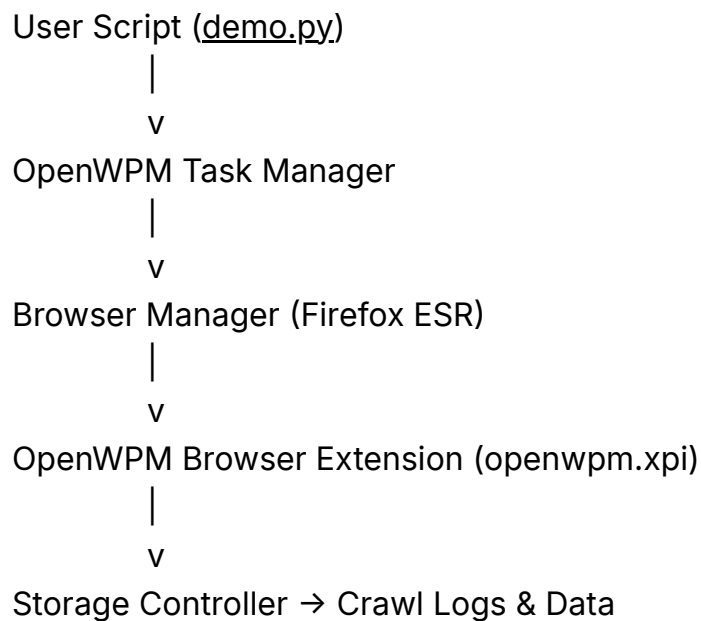
Software	Requirement
OS	Kali Linux (Debian-based)
Python	Miniconda (Conda environment)
Browser	Firefox ESR
Node.js	Required for extension build
npm	Required for extension build
Git	Clone repository
Build Tools	gcc, make, build-essential
Virtualization (optional)	Oracle VirtualBox

5. Architecture Overview

OpenWPM architecture consists of the following components :

- **Task Manager** : Orchestrates crawl tasks
- **Browser Manager** : Controls Firefox browser instances
- **Instrumentation Layer** : Injects measurement hooks
- **OpenWPM Extension (openwpm.xpi)** : Captures browser-side events
- **Storage Controller** : Stores logs, crawl data, and metadata
- **Crawler Scripts** : User-defined crawl logic

High-Level Flow :



6. Data Collected by OpenWPM

OpenWPM can collect :

- HTTP requests and responses
- First-party and third-party cookies
- Third-party domains and trackers
- JavaScript API calls
- Browser fingerprinting attributes
- LocalStorage / IndexedDB usage

- Redirect chains
- Page navigation events
- Link counts
- Browser profiles and crawl metadata

7. System Preparation

7.1 Update Operating System

```
sudo apt update  
sudo apt upgrade -y
```

7.2 Install Core Dependencies

```
sudo apt install -y \  
git wget curl build-essential gcc make \  
nodejs npm \  
python3-dev pkg-config \  
libgtk-3-0t64 libdbus-glib-1-2 libxt6t64 xvfb \  
ca-certificates unzip \  
libnss3 libasound2t64 libx11-xcb1 libxcomposite1 \  
libxdamage1 libxrandr2 libgbm1 libpangocairo-1.0-0
```

8. Install Miniconda

8.1 Download Installer

```
cd ~  
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_  
64.sh
```

8.2 Install

```
bash Miniconda3-latest-Linux-x86_64.sh
```

During installation :

- Press `Enter` to review license
- Type `yes` to accept license
- Press `Enter` to confirm install location
- Type `yes` to initialize Miniconda

8.3 Reload Shell

```
source ~/.bashrc
```

Verify :

```
conda --version
```

Expected Output :

```
conda x.x.x
```

9. Clone OpenWPM Repository

```
cd ~  
git clone https://github.com/openwpm/OpenWPM.git  
cd OpenWPM
```

10. OpenWPM Installation

```
./install.sh
```

This Step :

- Creates the `openwpm` Conda environment
- Installs Python dependencies
- Sets up OpenWPM modules

Activate environment :

```
conda activate openwpm  
which python
```

11. Firefox Installation & Configuration (Kali-Compatible)

Install Firefox ESR :

```
sudo apt update  
sudo apt install -y firefox-esr
```

Verify :

```
firefox-esr --version
```

Set Firefox binary :

```
export FIREFOX_BINARY="$(which firefox-esr)"
```

(Optionally add to `~/.zshrc` for persistence.)

12. OpenWPM Browser Extension Build

```
cd ~/OpenWPM/Extension  
npm install  
npm run build
```

Verify Extension :

```
ls ~/OpenWPM/Extension/openwpm.xpi
```

13. Validation – Running OpenWPM Demo

```
cd ~/OpenWPM  
python demo.py
```

Headless (optional) :

```
xvfb-run -a python demo.py
```

14. Execution Results

- Firefox ESR launched successfully
- OpenWPM extension loaded
- Demo websites crawled
- Link counting and command sequence completed
- Storage controller initialized and shut down cleanly

This confirms OpenWPM is operational.

15. Troubleshooting

Issue	Cause	Resolution
Firefox binary fails	Incompatible manual binary	Use <code>firefox-esr</code>
<code>openwpm.xpi</code> missing	Extension not built	Run <code>npm install</code> + <code>npm run build</code>
Browser crashes	Wrong FIREFOX_BINARY	Re-export correct path
Headless issues	Display not available	Use <code>xvfb-run</code>

16. Reproducibility Checklist

-  System packages installed

-  Conda environment created and activated
-  OpenWPM installed
-  Firefox ESR configured
-  Extension built (`openwpm.xpi`)
-  Demo executed successfully

17. Conclusion

OpenWPM has been successfully deployed and validated on Kali Linux. The tool is now ready for **automated privacy measurement, web tracking analysis, and large-scale crawling**. This document can be used as a **standard operating procedure (SOP)** for future deployments.